DECCAN FOODS

Customer Mobile App

Technical Design Document

Shopify-connected customer login, weekly baskets, subscriptions and delivery

Document	Value
Version	1.0
Date	16 September 2026
Prepared for	Deccan Foods
Channels	iOS, Android and Shopify web
Launch market	India first; UK/EU-ready architecture
Status	Build-ready baseline for discovery, estimation and implementation

Architecture decision

Build a Deccan-branded React Native mobile app connected to the existing Shopify store. Shopify remains the master for products, prices, inventory, discounts, checkout, payments, customers and orders. A small Deccan backend manages weekly-basket rules, delivery slots, notifications, loyalty and safe subscription actions.

deccanfood.com

1. Executive decision

Recommendation: connect the app to Shopify

Yes. Do not recreate ecommerce from scratch. The existing Shopify store already contains Deccan product data and order capability. The app should use Shopify Storefront API for catalogue/cart, Customer Account API for login/account, Shopify-hosted checkout through Checkout Kit, and a subscription solution based on Shopify Selling Plans and Subscription Contracts.

Decision area	Selected approach	Why
Mobile technology	React Native + TypeScript	One codebase for iOS and Android; strong Shopify integration path.
Commerce core	Existing Shopify store	One catalogue, inventory, order, discount and payment truth across web and app.
Login	Shopify Customer Account API using OAuth 2.0 / PKCE	Shared customer identity and order history; no duplicate passwords.
Checkout	Shopify Checkout Kit / hosted checkout	Preserves Shopify payment methods, tax, shipping, discounts and compliance.
Subscriptions	Shopify selling plans + established subscription app for MVP	Lower launch risk than writing a payment/subscription engine.
Custom backend	Node.js/TypeScript BFF + PostgreSQL	Protects secrets and supports baskets, slots, preferences, loyalty and webhooks.
Notifications	Firebase Cloud Messaging + APNs	Order, renewal, basket-cut-off and delivery notifications.
Analytics	Shopify analytics + Firebase/GA4	Commercial reporting and app funnel visibility.

1.1 MVP outcome

A customer can install the app, sign in to the same account used on deccanfood.com, browse products, create or choose a weekly basket, select one-time purchase or recurring delivery, pay through Shopify checkout, manage upcoming deliveries, and receive updates. The Shopify website and app show consistent products, pricing, stock and order history.

1.2 Scope boundaries

In MVP	After MVP
Customer registration/login; catalogue; search; favourites	AI meal planning and personalised nutrition
Pre-built weekly baskets and configurable basket within rules	Open marketplace and third-party supplier portal
Weekly/fortnightly/monthly subscription options	Complex household profiles and multi-address schedules
Skip, pause, resume and cancel subscription	Wholesale/B2B ordering and credit terms
Shopify checkout, discounts, addresses and order history	Own payment gateway or non-Shopify order engine
Delivery slots, push notifications and support	Route optimisation and driver application

2. Current-state observations

The live domain is served by Shopify and currently publishes Deccan Foods products including sauces, state-inspired biscuits and soft fruit bars. The app design therefore treats Shopify product/variant IDs, availability, prices and media as authoritative. At implementation start, the team must confirm the myshopify.com store domain, Shopify plan, markets, payment gateway, fulfilment locations and installed subscription/bundle apps.

Finding	Design implication
Shopify storefront is live at deccanfood.com	Use the existing shop as system of record; no catalogue migration for MVP.
Products have variants, tags, types, images and availability	Map app categories and filters from collections, tags and metafields.
Some products may be unavailable during catalogue setup	App must respect availableForSale and inventory; never accept an unavailable line.
India localisation is active	Use INR, Indian addresses, GST-ready order data and PIN-code serviceability.
Product portfolio is expanding	App navigation must be collection-driven rather than hard-coded.

3. Product vision and users

Persona	Goal	Key journeys
New customer	Discover trusted Deccan products and place first order	Browse > product > basket > checkout > delivery updates
Regular household	Receive a predictable weekly food basket	Subscribe > customise > pay > skip/swap > receive
Existing Shopify customer	Use existing identity and see past orders	Login > account > order history > reorder
Gift buyer	Send a curated basket to another address	Choose gift > message > recipient address > checkout
Customer care	Resolve order/subscription issues quickly	Search customer > order timeline > refund/replace/escalate
Commerce admin	Manage products, prices, inventory and promotions once	Shopify Admin > publish > app/web synchronise

4. Functional requirements

ID	Requirement	Priority	Acceptance summary
FR-01	Customer registration and login	Must	Email-based Shopify customer authentication; secure logout; session refresh.
FR-02	Shop and search	Must	Collections, products, variants, price, dietary tags, availability and search.
FR-03	Product details	Must	Images, ingredients, allergens, nutrition, pack size, storage, delivery and subscription options.
FR-04	Cart	Must	Add/update/remove lines, discount codes, delivery estimate and persistent cart.
FR-05	Weekly basket	Must	Pre-built basket plus swap/customise within price, category, weight and stock rules.
FR-06	Subscription purchase	Must	Weekly, fortnightly or monthly selling plan; clear price and renewal terms.
FR-07	Subscription management	Must	View next date; skip; pause; resume; permitted swaps; cancel with reason.
FR-08	Checkout/payment	Must	Shopify checkout; supported payment methods; address, shipping, tax and discounts.
FR-09	Orders	Must	Order history, status timeline, tracking, invoice link and reorder.
FR-10	Delivery slots	Must	PIN-code eligibility, available day/window, capacity and cut-off validation.
FR-11	Notifications	Must	Order confirmation, renewal reminder, edit cut-off, dispatch, delivery and failure.
FR-12	Favourites and reorder	Should	Save products and recreate an eligible prior cart.
FR-13	Support	Should	FAQ, WhatsApp/call/email entry and contextual issue submission.
FR-14	Loyalty/referrals	Later	Earn/redeem points and referral rewards after core ordering stabilises.
FR-15	Multi-market	Later	India and UK catalogues, currency, delivery, tax and compliance by Shopify Market.

4.1 Weekly basket business rules

Rule	MVP behaviour
Basket types	Essentials, Breakfast, Healthy Snacks, Family, Five-State Discovery, Build Your Own.
Frequency	One-time, weekly, every two weeks, or every four weeks.

Rule	MVP behaviour
Minimum	Configured per basket in metafields; default example is 5 eligible units or a minimum basket value.
Swaps	Only within the same swap group or within permitted price variance; stock validated before save and renewal.
Cut-off	Customer changes close at a configurable time before fulfilment; show exact local date/time.
Out of stock	Auto-substitute only with prior consent; otherwise notify, request choice, issue credit or skip affected line.
Price change	Notify before renewal where required; display current expected total before confirmation.
Address/slot	One active delivery address and slot per scheduled fulfilment; validate PIN code and capacity.
Cancellation	Self-service, clear effective date, reason capture; no dark patterns.
Food information	Ingredients, allergens, nutrition, storage, use-by/best-before policy and manufacturer details visible.

4.2 Customer journeys

Journey	Happy path
New subscriber	Open app > choose location > browse weekly baskets > select basket > customise > choose frequency/day > sign in > Shopify checkout > confirmation.
Existing customer	Open app > sign in with Shopify > view active subscription and orders > reorder or edit next basket.
Skip delivery	Subscription > upcoming delivery > Skip > confirmation > backend updates contract/schedule > push/email confirmation.
Swap item	Upcoming basket > eligible item > swap group > select replacement > price/stock recalculated > confirm before cut-off.
Payment failure	Webhook received > mark attention required > notify customer > secure payment-update/customer portal path > retry policy.
Out-of-stock renewal	Inventory event/check > apply consented substitute or request choice > preserve audit trail > fulfil/credit/skip.

5. UX and information architecture

Bottom navigation	Primary screens
Home	Greeting, delivery location, next basket, promotions, categories, recommended products
Shop	Collections, search, filters, product list, product detail
Basket	Cart, weekly basket builder, subscription selector, delivery estimate
Deliveries	Upcoming deliveries, calendar, skip/swap/pause, tracking
Account	Profile, addresses, orders, payments portal, preferences, support, legal

5.1 Required screens

Area	Screens
Onboarding	Splash; location/PIN code; value proposition; notification consent; sign in/create account
Commerce	Home; collection; search; filter; product detail; favourites; cart; checkout hand-off
Basket	Basket catalogue; basket detail; customise; frequency; delivery day/slot; review
Subscription	Subscription list; detail; next basket; swap; skip; pause; resume; cancel; history
Order	Confirmation; orders; order detail; delivery tracking; invoice; support issue
Account	Profile; addresses; communication preferences; privacy; logout; account deletion request
State/error	Offline; maintenance; empty; no stock; no service area; payment failed; session expired

5.2 Design requirements

- Use the Deccan dark-green, gold and cream identity with strong product photography and accessible contrast.
- Design mobile first for 360-430 px widths; support dynamic text and screen readers.
- Always show one-time price and subscription price/frequency separately; disclose renewal and cancellation terms before payment.
- Show ingredients and allergens above marketing copy for food safety.
- Keep Shopify checkout visually continuous using brand settings, return URLs and app deep links.
- Use skeleton loading, optimistic cart updates with rollback, and clear stock/price refresh messages.

6. Solution architecture

Layer	Components	Responsibility
Mobile client	React Native, TypeScript, React Navigation, TanStack Query, secure storage	UI, local state, catalogue/cart actions, login redirect, Checkout Kit, push token.

Layer	Components	Responsibility
Commerce APIs	Shopify Storefront API, Customer Account API, Checkout Kit	Products, collections, selling plans, carts, customer identity/account, checkout.
Deccan BFF	Node.js/NestJS or Fastify, GraphQL/REST, OAuth validation	App-safe orchestration, basket rules, slots, subscription actions, rate limiting.
Operational data	PostgreSQL; Redis optional	Delivery capacity, app preferences, device tokens, idempotency, audit events; no card data.
Shopify admin	GraphQL Admin API, webhooks, subscription app	Orders, fulfilment, inventory, subscription contracts and authoritative admin mutations.
Messaging	FCM/APNs, email provider, WhatsApp optional	Transactional and service notifications with consent controls.
Observability	Sentry/Application Insights, structured logs, uptime checks	Crash reporting, API tracing, alerting and service health.
Admin operations	Shopify Admin + small Deccan operations console	Commerce administration in Shopify; basket/slot/support controls in Deccan console.

6.1 Logical flow

1. Customer app	2. Deccan/Shopify services	3. Systems of record
Browse and build basket	Storefront API + basket rules BFF	Shopify catalogue + Deccan basket configuration
Authenticate	Customer Account OAuth/PKCE	Shopify customer identity
Pay	Checkout Kit opens Shopify checkout	Shopify order/payment
Manage subscription	BFF validates customer and request	Subscription app/Shopify contract
Receive updates	Shopify webhook > BFF > push/email	Event log + Shopify order/subscription

Data ownership rule
Shopify owns commerce truth. The Deccan database may cache Shopify identifiers and read models, but it must not become a competing master for price, stock, customer identity, payment or order status.

7. Shopify integration design

Capability	Shopify interface	Implementation
Catalogue	Storefront GraphQL API	Query collections/products/variants/metafields; cursor pagination; cache public data briefly.
Cart	Storefront Cart API	Create cart, add/update/remove lines, buyer identity, discount codes and delivery data.
Selling plans	Storefront API	Read sellingPlanAllocations and attach sellingPlanId to recurring cart lines.
Authentication	Customer Account API	Authorization Code + PKCE; app-link redirect; tokens stored in OS secure storage.
Customer account	Customer Account GraphQL API	Profile, addresses and order data within granted scopes.
Checkout	Checkout Kit	Present Shopify-hosted checkout; observe completion/cancel; use universal/app links.
Admin actions	GraphQL Admin API via backend only	Orders, fulfilment context, inventory/subscription operations; least privilege.
Events	Shopify webhooks	HTTPS endpoint; verify HMAC; queue; deduplicate; process asynchronously; replay failures.
Subscription engine	Selling Plans + Subscription Contracts	Use a Shopify subscription app in MVP; expose safe actions through provider APIs/portal or BFF.
Content extension	Metafields/metaobjects	Nutrition, allergens, dietary flags, state, basket eligibility, swap group and shelf/storage data.

7.1 Required Shopify configuration

1. Confirm the permanent {store}.myshopify.com domain and add the Headless sales channel or appropriate custom-app configuration.
2. Create Storefront API access and select a supported, pinned quarterly API version. Review upgrades before expiry.
3. Enable new customer accounts and configure Customer Account API OAuth clients, scopes, redirect URIs, logout URI and app links.
4. Configure Checkout Kit and branded checkout; test payment, discounts, tax, shipping, cancellation and return-to-app flows.
5. Install/select a subscription app that supports the target gateway, India operations, selling plans, customer self-service and usable APIs.
6. Create Shopify collections and metafield definitions for app navigation, food data and weekly-basket rules.
7. Register HTTPS webhooks, webhook HMAC secret, retry/dead-letter processing and monitoring.
8. Create app-specific discount rules and confirm compatibility with subscription selling plans and bundles.

7.2 Recommended metafields/metaobjects

Namespace.key	Type	Purpose
deccan.ingredients	Rich text / list	Ingredients presented consistently in app and web.
deccan.allergens	List of text/reference	Allergen declaration and filtering.

Namespace.key	Type	Purpose
deccan.dietary_flags	List	Vegetarian, vegan, gluten-free claim status, etc.; only verified claims.
deccan.state_origin	Metaobject reference	Karnataka, Kerala, Tamil Nadu, Andhra Pradesh, Telangana.
deccan.basket_eligible	Boolean	Eligibility for Build Your Own.
deccan.swap_group	Text/reference	Permitted substitution family.
deccan.units	Integer/decimal	Basket unit/weight contribution.
deccan.storage	Rich text	Storage and after-opening instructions.
deccan.delivery_class	Text	Ambient, chilled, fragile or restricted handling.
deccan.subscription_badge	Text	Customer-facing benefit/badge copy.

7.3 Webhook topics

Final topic names must be validated against the selected Shopify API version and subscription provider. At minimum process order create/update/cancel, fulfilment events, refund events, product/variant and inventory changes, customer updates/deletions, subscription contract/billing changes where available, and mandatory privacy webhooks. Webhook processing is at-least-once: verify, persist event ID, acknowledge quickly, then process idempotently from a queue.

8. Subscription design

MVP strategy
Use an established Shopify subscription app and Shopify-native selling plans. Build a Deccan-branded management experience only where the provider exposes supported APIs. Where an action is not safely available, open the provider/customer account portal in an authenticated web view. Do not directly store card details or invent recurring charges.

Object	Owner	Notes
Selling plan group	Shopify/subscription app	Weekly, fortnightly, four-weekly plans and discounts.
Subscription contract	Shopify/subscription app	Customer, lines, schedule, delivery and payment linkage.
Weekly basket template	Deccan backend + Shopify references	Name, eligible lines, min/max, swap groups, cut-off and artwork.
Upcoming basket snapshot	Deccan backend	Editable view linked to subscription/next order; versioned and audited.
Payment method	Shopify/payment gateway	Never stored in Deccan database.
Subscription action audit	Deccan backend	Who/when/action/old/new/provider result/idempotency key.

8.1 Subscription action rules

Action	Validation	Result
Subscribe	Plan available; products eligible; address serviceable; customer consent captured	Selling-plan cart checked out and contract created.
Skip	Before cut-off; next order not already generated/locked	One cycle skipped; new date and confirmation returned.
Pause	No active fulfilment conflict; policy allows pause	Future billing/fulfilment paused until chosen date or resume.
Swap	Eligible swap group; price/stock/dietary checks	Upcoming basket and provider contract lines updated atomically where supported.
Cancel	Identity revalidated; effective date shown	Contract cancelled according to policy; confirmation and audit event.
Payment update	Secure provider flow only	Customer completes hosted flow; webhook confirms status.

9. Backend API design

Endpoint	Method	Purpose
/v1/config	GET	Market, feature flags, minimum versions and support contacts.
/v1/serviceability/{postcode}	GET	Check delivery area, days, fees and cut-off.
/v1/baskets	GET	Available basket templates for market and customer.
/v1/baskets/{id}/validate	POST	Validate selected lines, totals, stock and rules.
/v1/delivery-slots	GET	Return capacity-aware slots for postcode/date.
/v1/subscriptions	GET	Return customer subscriptions and next delivery read model.
/v1/subscriptions/{id}/skip	POST	Skip next eligible cycle with idempotency.
/v1/subscriptions/{id}/pause	POST	Pause according to policy/provider capability.
/v1/subscriptions/{id}/resume	POST	Resume and return next date.
/v1/subscriptions/{id}/basket	PUT	Validate and update upcoming basket.
/v1/subscriptions/{id}/cancel	POST	Cancel with reason and explicit confirmation.
/v1/devices	POST/DELETE	Register/remove push token and preferences.
/v1/support/cases	POST	Create contextual support request.
/webhooks/shopify	POST	Verify and enqueue Shopify webhooks.
/webhooks/subscriptions	POST	Verify and enqueue provider events.

9.1 API conventions

- JSON over HTTPS; OpenAPI 3.1 contract; /v1 versioning; UTC timestamps with market-local presentation.
- Bearer token derived from validated Shopify customer identity; backend maps subject to Shopify customer ID.

- Idempotency-Key required for all state-changing subscription, slot and support operations.
- Structured errors: code, customer-safe message, correlationId, retryable and fieldErrors.
- Server enforces ownership; a customer can never act on another customer ID supplied by the client.
- Rate limit by customer/device/IP; protect login, postcode, basket validation and mutation endpoints.

10. Data model

Entity	Key fields	Retention/ownership
customer_profile	shopify_customer_id, preferences, market, consent timestamps	Minimal app profile; Shopify owns identity/contact data.
device	customer_id, platform, push_token_hash, locale, last_seen	Delete on logout/token invalidation; protect raw token.
basket_template	id, name, market, rules_json, active, version	Deccan master; Shopify variant references.
basket_selection	customer/subscription, template_version, line refs, state, cut_off	Retain operational/audit period; no card data.
delivery_zone	postcode pattern, days, fee, minimum, active	Deccan operations master.
delivery_slot	zone, start/end, capacity, reserved, cut_off	Transactional with concurrency control.
subscription_read_model	provider_id, shopify_contract_id, status, next_date	Webhook-derived cache; provider remains authoritative.
webhook_event	provider, event_id, topic, received_at, status, attempts	Payload minimised/encrypted; idempotency and audit.
audit_event	actor, action, entity, before/after hash, time, correlation	Append-only security/operations trail.
support_case	customer, order/subscription ref, category, message, status	Retention based on support/privacy policy.

11. Security and privacy

Control	Requirement
Authentication	Customer Account OAuth Authorization Code with PKCE; short-lived access tokens; secure refresh/session handling.
Mobile secrets	No Admin API token, private Storefront token or webhook secret in the app bundle. Use OS Keychain/Keystore for tokens.
Authorisation	Backend derives customer identity from validated token and applies object-level ownership checks.
Payments	Shopify-hosted checkout/payment update. Deccan systems never collect or store full card or UPI credentials.
API security	TLS, least-privilege scopes, schema validation, rate limiting, WAF/API gateway, dependency scanning.
Webhook security	Raw-body HMAC verification, replay/idempotency protection, fast acknowledgement and queued handling.
Data protection	Encryption in transit/at rest; field minimisation; secret manager; production access logging and reviews.
Privacy	Consent records; export/deletion workflow; Shopify mandatory privacy webhooks; retention schedule.
App security	Root/jailbreak signals as risk input, certificate pinning only with safe rotation, no sensitive logs, release signing.

Control	Requirement
Secure delivery	SAST, dependency/SBOM scanning, secret scanning, DAST/API tests, penetration test before public launch.
Compliance	Confirm India DPDP obligations, consumer/subscription terms, GST/invoicing, food labelling and target-market rules with advisers.
Incident response	Severity model, alerting, token/secret rotation, customer communication path and tested recovery runbook.

12. Non-functional requirements

ID	Area	Target
NFR-01	Availability	99.9% monthly for Deccan API in production, excluding announced maintenance and Shopify/provider outages.
NFR-02	Performance	Cached home/catalogue response p95 < 1.5 s on typical 4G; BFF p95 < 500 ms excluding dependencies.
NFR-03	Reliability	No duplicate mutation from retry; webhook and subscription actions idempotent.
NFR-04	Scalability	Stateless API horizontally scalable; queue absorbs event spikes; DB indexes reviewed from telemetry.
NFR-05	Accessibility	WCAG 2.2 AA-oriented UI; screen reader labels, contrast, touch targets and dynamic text.
NFR-06	Offline	Graceful cached browsing; cart mutations queued only when safe; checkout/subscription changes require network.
NFR-07	Observability	Crash-free sessions >= 99.5%; correlation across app/API/webhook; PII-redacted logs.
NFR-08	Compatibility	Current and two prior major iOS versions; Android support baseline agreed from customer analytics.
NFR-09	Recovery	Initial RPO <= 15 min and RTO <= 4 h for Deccan operational DB; tested backups.
NFR-10	Maintainability	Typed API contracts, automated tests, feature flags and controlled Shopify API upgrades.

13. Notifications and communications

Trigger	Channel	Timing/content
Order confirmed	Push + email	Immediately; order number, total and delivery estimate.
Edit window opens	Push	Upcoming basket available for customisation.
Edit cut-off reminder	Push + optional email	24-48 hours before cut-off; exact deadline.

Trigger	Channel	Timing/content
Renewal/billing reminder	Email/push as policy requires	Amount, date, basket and manage link.
Payment failed	Push + email	Clear action to secure hosted payment update; do not include sensitive data.
Dispatched/out for delivery	Push	Tracking and delivery window.
Substitution required	Push + email	Choice deadline and acceptable alternatives.
Paused/cancelled/skipped	Push + email	Effective date and next status for audit confidence.

14. Analytics and KPIs

Funnel/operation	Events and measures
Acquisition	install, first_open, source/campaign, permission outcomes
Discovery	home_view, search, filter, product_view, favourite, no_results
Basket	basket_start, item_add/swap/remove, validation_error, basket_complete
Checkout	checkout_start, checkout_complete, checkout_cancel, discount_applied
Subscription	plan_view, subscribe, skip, pause, resume, cancel, failed_payment, churn reason
Delivery	slot_view/select, fulfilment delay, delivered, issue reported
Commercial KPIs	conversion, AOV, subscription attach rate, active subscribers, churn, retention, LTV, repeat rate
Operational KPIs	fill rate, substitution rate, on-time delivery, payment recovery, webhook lag, support contacts/order

Analytics must not include passwords, access tokens, payment data, full addresses or free-text support content. Define a versioned event dictionary before implementation.

15. Testing strategy

Level	Coverage
Unit	Basket rule engine, price displays, date/cut-off calculations, validation, reducers and error mapping.
Contract	GraphQL queries, BFF OpenAPI, webhook payloads and subscription-provider adapters.
Integration	Shopify development store, test products/selling plans, customer OAuth, checkout and provider sandbox.
End to end	New/existing user, one-time order, subscription, skip/pause/swap/cancel, payment failure and out-of-stock.
Mobile	iOS/Android matrix, deep links, push, accessibility, offline/poor network, upgrade and background/foreground.
Performance	Home/catalogue load, cart concurrency, slot contention, webhook burst and API dependency latency.

Level	Coverage
Security	Threat model, authz tests, token/log review, dependency scan, API penetration test and webhook replay.
UAT	Commerce, operations, customer service and finance sign-off using production-like products and policies.

15.1 MVP acceptance criteria

- The same published Shopify products, current prices and availability appear on website and app within the agreed cache window.
- An existing website customer can authenticate in the app and see eligible order history without creating a duplicate account.
- One-time and recurring baskets complete through Shopify checkout and create correct Shopify orders/contracts.
- Skip, pause, resume, cancellation and supported swaps are reflected in the provider/Shopify and confirmed to the customer.
- Duplicate API/webhook delivery cannot create duplicate actions or customer messages.
- No Shopify private/admin secret or payment data is present in app builds, analytics or logs.
- Critical accessibility, security and crash defects are closed before store submission.

16. DevOps and environments

Environment	Purpose	Rules
Local	Developer implementation	Mock services or dedicated dev store; no production customer data.
Development	Shared integration	Shopify dev store, test subscription/payment, synthetic data.
Staging	UAT and release candidate	Production-like configuration; separate keys/database; controlled testers.
Production	Public app	Protected branches, approvals, secret manager, monitoring, backups and incident alerts.

- CI: lint, typecheck, unit tests, API contract tests, dependency/secret scan and signed build.
- CD: backend blue/green or rolling deployment; database migrations backward-compatible; mobile staged rollout.
- Release controls: feature flags, minimum supported version, remote kill switch for risky subscription functions.
- Branching: short-lived branches, reviewed pull requests and tagged releases; infrastructure as code.

17. Delivery plan and estimate

Phase	Indicative duration	Outputs
0. Discovery and Shopify audit	1 week	Store/plan/app audit, customer policies, subscription provider decision, backlog and UX map.
1. Foundation and UX	2 weeks	Design system, navigation, API contracts, environments, CI/CD and Shopify dev configuration.
2. Commerce MVP	3 weeks	Login, catalogue, search, product, cart, checkout, orders and analytics.
3. Weekly baskets/subscriptions	3 weeks	Basket builder, selling plans, manage actions, delivery slots, webhooks and notifications.
4. Hardening and UAT	2 weeks	E2E/security/performance/accessibility, operations training and store assets.
5. App review and rollout	1-2 weeks	TestFlight/internal testing, submissions, staged launch and hypercare.

Planning range
A focused MVP is approximately 10-13 weeks with a small experienced team, subject to subscription provider capability, Shopify plan/payment setup, final basket rules, content readiness and app-store review. Start with pre-built weekly baskets; release fully flexible recurring swaps only after provider API validation.

17.1 Suggested team

Role	Indicative involvement
Product owner / Deccan decision-maker	Policies, priorities, UAT and commercial sign-off
UX/UI designer	Mobile flows, design system, prototype and accessibility
React Native engineers (1-2)	iOS/Android app and native integrations
Backend/Shopify engineer	BFF, Admin API, webhooks, subscription/provider integration
QA engineer	Automation, device matrix, regression and UAT support
DevOps/security support	Environments, delivery, monitoring, hardening and review

18. Risks and mitigations

Risk	Impact	Mitigation
Subscription provider lacks required customer actions/API	High	Provider proof-of-concept in week 1; retain hosted portal fallback; phase advanced swaps.
Duplicate customer identities between web and app	High	Use Shopify Customer Account API; no parallel auth/customer database.
Stock changes between basket build and renewal	High	Validate at edit, cut-off and renewal; consent-based substitution workflow.

Risk	Impact	Mitigation
Checkout/payment method mismatch in India	High	Test selected gateway, UPI/COD policy and recurring-payment support before build commitment.
Bundle/subscription/discount incompatibility	High	Create matrix and test actual Shopify apps/plans in a development store.
App rejected for unclear subscription terms	Medium	Prominent recurring price, frequency, cancellation and policy links; physical goods use Shopify checkout.
API version or provider change	Medium	Pin API versions, adapter layer, quarterly upgrade review and contract tests.
Food/allergen content incomplete	High	Make verified content a publication gate; do not infer claims from product names.
Delivery capacity oversold	Medium	Transactional slot reservations, expiry, reconciliation and admin override.
Excess custom scope delays launch	High	MVP guardrails, feature flags and staged roadmap.

19. Decisions required before build

#	Decision	Owner
1	India-only MVP or simultaneous UK launch?	Founder/Product
2	Pre-built baskets only, or build-your-own at MVP?	Product/Operations
3	Allowed frequencies, cut-off, cancellation, substitutions and refunds?	Commercial/Legal
4	Selected Shopify subscription and bundle apps after technical proof?	Product/Engineering
5	Payment gateway and recurring-payment support for target market?	Finance/Shopify
6	Delivery zones, slots, fees, minimum value and fulfilment partner?	Operations
7	Verified product ingredients, allergens, nutrition and storage content?	Quality/Product
8	Loyalty/referral in MVP or later?	Commercial
9	Hosting choice and support ownership?	Engineering
10	Final privacy, subscription, shipping, returns and cancellation wording?	Legal/Product

20. Build hand-off checklist

- Shopify staff/development access and permanent shop domain
- Installed apps, Shopify plan, Markets, locations, tax and payment configuration
- Brand assets, app icon, splash, fonts, colours and product imagery
- Customer-account OAuth client and approved redirect/app-link domains
- Subscription/bundle provider credentials and confirmed API capability

- Product data completion, collection structure and metafield definitions
- Delivery zones/slots/cut-offs, fulfilment workflow and notification templates
- Apple Developer and Google Play Console accounts; privacy policy/support URL
- Analytics, monitoring, support and incident-response accounts
- UAT users, devices, scenarios and launch acceptance authority

Appendix A. Example cart and basket payloads

Illustrative app-to-BFF request (not a final API contract):

```
POST /v1/baskets/basket_family_weekly/validate
Authorization: Bearer <customer-access-token>
Idempotency-Key: 3f9c...

{
  "market": "IN",
  "frequency": "WEEKLY",
  "deliveryPostcode": "560100",
  "requestedSlotId": "fri-am-zone-ecity",
  "lines": [
    {"shopifyVariantId": "gid://shopify/ProductVariant/...", "quantity": 2},
    {"shopifyVariantId": "gid://shopify/ProductVariant/...", "quantity": 3}
  ]
}
```

Appendix B. Definition of done

Dimension	Done when
Product	Acceptance criteria approved; policies and edge cases implemented; analytics instrumented.
Engineering	Reviewed code merged; tests pass; migrations/runbooks complete; feature flag configured.
Security	Secrets/scopes reviewed; no critical/high unresolved findings; logs checked for PII/tokens.
Quality	Device/browser matrix passed; accessibility reviewed; regression complete.
Operations	Shopify/support/fulfilment teams trained; alerts and dashboards active.
Release	Store metadata/privacy disclosures approved; rollback/kill switch tested; owner signs off.

Appendix C. Reference sources

Deccan Foods live storefront: https://www.deccanfood.com/

Shopify Storefront API: https://shopify.dev/docs/api/storefront/latest

Shopify Customer Account API: https://shopify.dev/docs/api/customer/latest

Shopify subscriptions overview: https://shopify.dev/docs/apps/build/purchase-options/subscriptions

Shopify webhooks: https://shopify.dev/docs/apps/build/webhooks

Shopify Checkout Kit for mobile: https://shopify.dev/docs/storefronts/headless/mobile-apps/checkout-kit

Technical references should be revalidated against the Shopify API version selected at implementation start; Shopify publishes versioned APIs and platform capabilities evolve.